

De novo assembly of *Roseomonas mucosa* isolated from patients with atopic dermatitis

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ABSTRACT *Roseomonas mucosa* is associated with the normal skin microflora. Here, we present *de novo* sequence assemblies from *R. mucosa* isolates obtained from the skin lesions of three atopic dermatitis patients.

KEYWORDS *Roseomonas*, genome sequence, atopic dermatitis

Roseomonas mucosa is a Gram-negative coccobacillus found in aquatic environments and from various clinical samples (1–5). *R. mucosa* is commonly isolated from the microbiota of human skin (4, 6). Atopic dermatitis (AD), an inflammatory skin disease, causes susceptibility to *Staphylococcus aureus* infection, immune dysregulation function, and an impaired skin barrier. Improved AD clinical outcome was observed when *R. mucosa* was used to topically treat patients with AD (2, 4, 7). This treatment had an improved outcome in mouse models of AD. Interestingly, this mouse model treated with *R. mucosa* from AD patients, had no impact or a worse clinical outcome (2). We present the sequences from three AD-sourced *R. mucosa* isolates (2).

R. mucosa from lesions of three AD patients was isolated using a FloqSwab moistened in phosphate-buffered saline. Swabs were placed in Reasoner's 2A (R2A) broth containing amphotericin B and vancomycin and incubated for 48–72 h at 32°C. Cultures were spread on R2A agar and incubated as above to form single colonies (4). Bacteria, stored at –80°C, were spread on R2A agar and a single colony used to inoculate 100 mls of R2A broth, and grown at 32°C for 20–25 h (8). Bacteria were washed, resuspended, and heated to 65°C, followed by separate incubations with proteinase k, lysozyme, and SDS. Each sample was incubated with RNaseI for 30 min at 37°C, then combined with a 3/4 vol of saturated phenol solution, mixed for 10 min and centrifugated at 12,800 × *g* for 10 min. The aqueous phase was removed and treated with phenol:chloroform:isoamyl alcohol (25:24:1) until the white precipitate interface was absent. A chloroform-only extraction was performed, and aqueous solution was collected. DNA was precipitated with 3 M sodium acetate, washed 2× with 70% EtOH, and resuspended in Qiagen's EB buffer.

PacBio and Illumina sequencing libraries were generated from the same DNA using the 10 kb Template Preparation and Sequencing kit with Covaris g-tubes (sheared to >8,700 bp) and Illumina TruSeq Nano DNA kit using a Covaris M220 (sheared to ~550 bp), respectively (PacBio protocol 100-092-800-06 and TruSeq protocol 15041110 rev. D). Libraries were sequenced with PacBio Sequel and Illumina MiSeq (2 × 300 bp read length) platforms. Illumina reads were adapter trimmed using cutadapt (v.1.12) and trimmed and filtered for quality using FASTX-Toolkit (v.0.0.14). Bioinformatic analysis and construction of the genomes were performed using the following programs: CANU (v.1.0), Bowtie2 (v.2.2.4), pilon (v.1.16), and MIRA (v.4.0). All software was used with default parameters. The PacBio data were used to construct DNA scaffolds which

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TABLE 1 - *Roseomonas mucosa* genome properties

Isolate	Raw read	SRA accession	illumina	PacBio raw	PacBio	Genbank	Chromosome	No. of	Coverage	%GC	Plasmid	Genbank	Plasmid	No. of	Coverage	%GC	Total
Bioproject	reference	no.	raw reads	reads	N50	accession	size (bp)	chromosomal	(mean	depth)	sequences	accession no.	name	size (bp)	(mean depth)	genome	features
AD1	PRJNA386446	SRX19873381	16,467,886	163,970	8,494	CP025196.1	4,135,189	4,008	625	71	6	CP025195.1	p1-AD1	441,278	634	68.1	4,632
		SRX19873380										CP025190.1	p2-AD1	207,850	709	67.4	
		SRX19873379										CP025192.1	p3-AD1	171,049	1,198	68.5	
												CP025191.1	p4-AD1	137,855	596	67.7	
												CP025193.1	p5-AD1	97,477	945	66.4	
												CP025194.1	p6-AD1	50,633	855	65.9	
AD2	PRJNA386450	SRX19877186	15,601,309	41,438	6,667	CP025189.1	4,277,502	4,151	485	70.7	5	CP025188.1	p1-AD2	415,646	435	68.3	5,028
		SRX19877185										CP025187.1	p2-AD2	160,042	459	68.3	
		SRX19877184										CP025184.1	p3-AD2	119,544	808	65.3	
												CP025185.1	p4-AD2	69,338	811	66.5	
												CP025186.1	p5-AD2	8,942	202	66.9	
AD3	PRJNA498095	SRX19880322	3,850,453	1,688,207	4,200	CP034902.1	4,219,833	4,095	1,690	70.5	5	CP034903.1	p1-AD3	385,778	1,555	68.0	4,989
		SRX19880321										CP034905.1	p2-AD3	180,748	1,537	65.5	
												CP034904.1	p3-AD3	156,070	177	70.0	
												CP034906.1	p4-AD3	25,394	1,276	66.5	
												CP034907.1	p5-AD3	29,268	700	68.5	

were optimized using Illumina data. Igenbio Inc. annotated the genomes, while public annotation was compiled by the NCBI Prokaryotic Genome Annotation Pipeline (9).

The *R. mucosa* genomes consist of a single large circular chromosome (4.01–4.28 Mb) and five to six autonomously replicating plasmid sequences (8.9–506.3 kb) (Table 1). Genome properties and annotation statistics are listed in Table 1. These genomes may help determine the genes that are important for successful treatment of AD.

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DATA AVAILABILITY

The chromosome and plasmid sequences from the *Roseomonas mucosa* AD1, AD2, and AD3 strains have been deposited in GenBank under the accession numbers presented in Table 1. A NCBI RefSeq genome assembly was created for AD3 ([ASM2622269v1](#)). Raw read sequences can be accessed at the NCBI sequence read archive with the following Bioproject numbers: [PRJNA386446](#), [PRJNA386450](#), and [PRJNA498095](#).

ETHICS APPROVAL

The National Institute of Allergy and Infectious Disease IRB approved the associated clinical trial (NCT02262819) and human sample collection and processing. Participants

gave full consent to sample collection and provided written consent to the research protocol.

REFERENCES

1. Kim YK, Moon JS, Song KE, Lee WK. 2016. Two cases of bacteremia due to *Roseomonas mucosa*. *Ann Lab Med* 36:367–370. <https://doi.org/10.3343/alm.2016.36.4.367>
2. Myles IA, Earland NJ, Anderson ED, Moore IN, Kieh MD, Williams KW, Saleem A, Fontecilla NM, Welch PA, Darnell DA, Barnhart LA, Sun AA, Uzel G, Datta SK. 2018. First-in-human topical microbiome transplantation with *Roseomonas mucosa* for atopic dermatitis. *JCI Insight* 3:e120608. <https://doi.org/10.1172/jci.insight.120608>
3. Myles IA, Moore IN, Castillo CR, Datta SK. 2018. Differing virulence of healthy skin commensals in mouse models of infection. *Front Cell Infect Microbiol* 8:451. <https://doi.org/10.3389/fcimb.2018.00451>
4. Myles IA, Williams KW, Reckhow JD, Jammeh ML, Pincus NB, Sastalla I, Saleem D, Stone KD, Datta SK. 2016. Transplantation of human skin microbiota in models of atopic dermatitis. *JCI Insight* 1:e86955. <https://doi.org/10.1172/jci.insight.86955>
5. Romano-Bertrand S, Bourdier A, Aujoulat F, Michon A-L, Masnou A, Parer S, Marchandin H, Jumas-Bilak E. 2016. Skin microbiota is the main reservoir of *Roseomonas mucosa*, an emerging opportunistic pathogen so far assumed to be environmental. *Clin Microbiol Infect* 22:737. <https://doi.org/10.1016/j.cmi.2016.05.024>
6. Cosseau C, Romano-Bertrand S, Duplan H, Lucas O, Ingrassia I, Pigasse C, Roques C, Jumas-Bilak E. 2016. *Proteobacteria* from the human skin microbiota: species-level diversity and hypotheses. *One Health* 2:33–41. <https://doi.org/10.1016/j.onehlt.2016.02.002>
7. Liu X, Qin Y, Dong L, Han Z, Liu T, Tang Y, Yu Y, Ye J, Tao J, Zeng X, Feng J, Zhang X-Z. 2023. Living symbiotic bacteria-involved skin dressing to combat indigenous pathogens for microbiome-based biotherapy toward atopic dermatitis. *Bioact Mater* 21:253–266. <https://doi.org/10.1016/j.bioactmat.2022.08.019>
8. Myles IA, Reckhow JD, Williams KW, Sastalla I, Frank KM, Datta SK. 2016. A method for culturing Gram-negative skin microbiota. *BMC Microbiol* 16:60. <https://doi.org/10.1186/s12866-016-0684-9>
9. Tatusova T, DiCuccio M, Badretdin A, Chetvernin V, Nawrocki EP, Zaslavsky L, Lomsadze A, Pruitt KD, Borodovsky M, Ostell J. 2016. NCBI prokaryotic genome annotation pipeline. *Nucleic Acids Res* 44:6614–6624. <https://doi.org/10.1093/nar/gkw569>